

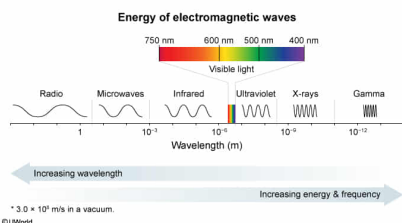
Which of the following expressions correctly compares the energies of visible light (E_{visible}), microwaves ($E_{\text{microwave}}$), and radio waves (E_{radio})?

- ☐ A. $E_{\text{radio}} = E_{\text{microwave}} < E_{\text{visible}}$ [2%]
- ✓ ☒ B. $E_{\text{radio}} < E_{\text{microwave}} < E_{\text{visible}}$ [78%]
- ☐ C. $E_{\text{microwave}} > E_{\text{radio}} > E_{\text{visible}}$ [16%]
- ☐ D. $E_{\text{radio}} = E_{\text{microwave}} > E_{\text{visible}}$ [2%]

Correct

77% answered correctly

Explanation:



Electromagnetic radiation refers to oscillating electric and magnetic fields that propagate radiant energy through a medium. The **electromagnetic spectrum** describes classifications of electromagnetic radiation that differ from one another in terms of waveform characteristics, occurrence in the natural world, and technological utility.

The electromagnetic spectrum is typically arranged in order of frequency (f) or wavelength (λ). Because the **frequency and wavelength** of electromagnetic radiation are **inversely proportional**, arranging the electromagnetic spectrum from largest wavelength (radio waves) to smallest wavelength (gamma rays) is equivalent to arranging the electromagnetic spectrum from smallest frequency to largest frequency.

Alternately, the electromagnetic spectrum may also be ordered in terms of the **energy (E)** associated with a given electromagnetic waveform, which can be expressed in terms of either electromagnetic frequency or wavelength using Planck's constant (h) and the speed of light (c) as:

$$E = h f = \frac{h c}{\lambda}$$

Accordingly, the energy of electromagnetic radiation is *inversely proportional* to wavelength and *directly proportional* to frequency. Consequently, the order of electromagnetic radiation types arranged from lowest energy to highest energy is:

E Radio waves

Therefore, radio waves have less energy than microwaves, and both these types of electromagnetic radiation have less energy than light visible to the human eye (**Choice B**).

Educational objective:

The electromagnetic spectrum consists of radiation types with different wavelengths, frequencies, and energies. Ordering the electromagnetic spectrum in terms of decreasing wavelength is equivalent to ordering the electromagnetic spectrum in terms of increasing energy.

Time spent: 0 sec
Subject:
Physics

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System:
Light and Sound